

Airline Seat Reservation System (ASRS)

A web-based airline seat reservation system built with PHP, MySQL, Bootstrap, and jQuery. This application allows users to book airline seats, manage reservations, and generate QR codes for boarding passes.

Table of Contents

- Features
 - Tech Stack
 - Project Structure
 - Installation
 - Database Schema
 - Usage
 - Screenshots
 - Security Considerations
 - Contributors
 - License
-

Features

User Features

- **User Registration & Login** - Secure authentication system with email/password
- **Profile Management** - Update personal information and upload KYC documents (PDF)
- **PNR-based Booking** - Link existing flight bookings using PNR numbers
- **Interactive Seat Selection** - Visual seat map with real-time availability
- **Payment & Ticketing** - View booking details with QR code generation
- **Booking History** - Track pending, completed, and boarded reservations

Airport Agent Features

- **Agent Login** - Separate authentication portal for airport staff
- **Booking Management** - View and manage passenger reservations

System Features

- **Responsive Design** - Mobile-friendly Bootstrap interface
 - **QR Code Generation** - Generate scannable QR codes for boarding passes
 - **Session Management** - Secure PHP session handling
 - **Print Ticket** - Printable ticket format
-

Tech Stack

Layer	Technology
Frontend	HTML5, CSS3, JavaScript, jQuery, Bootstrap 4.5
Backend	PHP 7+
Database	MySQL
Libraries	jQuery Seat Charts, QRCode.js, Font Awesome
Server	Apache/Nginx

Project Structure

```

ASRS/
├── about.php           # About Us page
├── bookings.php       # Pending bookings list
├── contact.php        # Contact information page
├── footer.php         # Common footer component
├── head.php           # Common head/meta tags component
├── headerasrs.php     # Navigation header component
├── index.html         # QR Code generator demo
├── layout.css         # Seat chart styling
├── layout.html        # Seat selection template
├── layout.js          # jQuery Seat Charts library
├── layout.php         # Main seat selection page
├── loginasrs.php      # Login page (User & Agent)
├── logout.php         # Session logout handler
├── main.php           # Homepage with carousel
├── message.php        # Alert/notification component
├── payment.php        # Payment & ticket details
├── pnrinfo.php        # PNR linking page
├── qrcode.js          # QR Code generation library
├── qrcode.min.js      # Minified QR Code library
├── qrcode.php         # PHP QR Code implementation
├── register.php       # User & Agent registration
├── rprofile.php       # User profile management
├── script.js          # Hamburger menu animation
├── Userpage.html      # User dashboard
├── bower.json         # Package configuration
├── jquery.min.js      # jQuery library
├── css/
│   ├── style.css      # Custom styles
│   └── styles.css     # Additional styles
├── image/
│   ├── ASR1.jpg - ASR12.jpg # Banner images
│   └── ASR6.png        # Logo/favicon
├── file/
│   ├── agentlogin.php  # Agent authentication
│   ├── agentreg.php    # Agent registration handler
│   ├── booking_connection.php # Database connection
│   └── bookind_add.php  # Add booking handler

```

```
| |— delete.php          # Delete booking handler
| |— depart_from_to.php # Flight details helper
| |— get_accomodation.php # Class info helper
| |— updateprofile.php  # Profile update handler
| |— userlogin.php      # User authentication
| |— userreg.php        # User registration handler
|— phpqrcode/          # PHP QR Code library
```

Installation

Prerequisites

- PHP 7.0 or higher
- MySQL 5.6 or higher
- Apache/Nginx web server
- Web browser with JavaScript enabled

Step-by-Step Setup

1. **Clone/Download** the project to your web server directory:

```
# For XAMPP
cp -r ASRS /opt/lampp/htdocs/

# For WAMP
cp -r ASRS C:/wamp64/www/
```

2. **Create the database:**

```
CREATE DATABASE airliness;
```

3. **Import database tables** (create the following tables):

- user - User accounts
- agent - Airport agent accounts
- passenger_details - Flight passenger information
- user_bookings - User-booking associations

4. **Configure database connection:** Edit file/booking_connection.php:

```
$servername = "localhost";
$username = "root";
$password = "your_password";
$dbname = "airliness";
```

5. **Set permissions:**

```
chmod 755 -R ASRS/
chmod 777 ASRS/images/ # For QR code generation
```

6. **Access the application:** Open <http://localhost/ASRS/main.php> in your browser.

Database Schema

User Table

```
CREATE TABLE user (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  uname VARCHAR(30) NOT NULL,  
  uemail VARCHAR(30) UNIQUE NOT NULL,  
  upassword VARCHAR(255) NOT NULL,  
  uphone VARCHAR(12) NOT NULL,  
  ucity VARCHAR(20) NOT NULL,  
  pdf_file VARCHAR(255)  
);
```

Agent Table

```
CREATE TABLE agent (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  aid VARCHAR(12) UNIQUE NOT NULL,  
  aname VARCHAR(30) NOT NULL,  
  aemail VARCHAR(30) UNIQUE NOT NULL,  
  apassword VARCHAR(255) NOT NULL,  
  aphone VARCHAR(12) NOT NULL,  
  acity VARCHAR(20) NOT NULL  
);
```

Passenger Details Table

```
CREATE TABLE passenger_details (  
  p_pnr VARCHAR(20) PRIMARY KEY,  
  p_name VARCHAR(50),  
  p_age INT,  
  p_sex VARCHAR(10),  
  p_fno VARCHAR(20),  
  p_from VARCHAR(50),  
  p_to VARCHAR(50),  
  p_dedate DATE,  
  p_ardate DATE,  
  p_detime TIME,  
  p_artime TIME,  
  p_status VARCHAR(20),  
  p_class VARCHAR(20),  
  p_passtype VARCHAR(20),  
  p_fid VARCHAR(20),  
  p_seats VARCHAR(50),  
  p_extra VARCHAR(100),  
  uid INT,  
  FOREIGN KEY (uid) REFERENCES user(id)  
);
```

User Bookings Table

```
CREATE TABLE user_bookings (  
    ubid INT AUTO_INCREMENT PRIMARY KEY,  
    uid INT,  
    p_pnr VARCHAR(20),  
    FOREIGN KEY (uid) REFERENCES user(id),  
    FOREIGN KEY (p_pnr) REFERENCES passenger_details(p_pnr)  
);
```

Usage

For Passengers

1. **Register** at `register.php` - Create a user account with KYC document upload
2. **Login** at `loginasrs.php` - Access your dashboard
3. **Link PNR** at `pnrinfo.php` - Add your flight booking using PNR number
4. **Select Seat** at `bookings.php` - Choose available seats from interactive map
5. **Payment** at `payment.php` - View ticket details and QR code
6. **Print Ticket** - Use browser print function for physical copy

For Airport Agents

1. **Register** as agent with personal ID
2. **Login** via agent tab
3. **Manage** passenger check-ins and boarding

Key Components

Seat Selection System (`layout.php`)

- Interactive seat map using jQuery Seat Charts
- Real-time seat availability (green = available, red = unavailable)
- Price calculation with cart functionality
- Keyboard navigation support (arrow keys, spacebar)

QR Code Generation (`payment.php`)

- Generates QR code containing PNR information
- Uses QRCode.js library for client-side generation
- Alternative PHP-based generation available

Authentication System

- Session-based authentication
 - Separate login portals for users and agents
 - Password validation with minimum length requirements
-

Security Considerations

⚠ Important: This is a student/academic project. The following security improvements are recommended for production use:

1. **SQL Injection** - Use prepared statements instead of string concatenation
2. **Password Hashing** - Implement `password_hash()` instead of plaintext storage
3. **XSS Protection** - Sanitize all user inputs with `htmlspecialchars()`
4. **CSRF Tokens** - Add CSRF protection to all forms
5. **File Upload** - Validate and restrict file types/sizes for KYC uploads
6. **HTTPS** - Enable SSL/TLS for all communications
7. **Input Validation** - Strengthen client-side and server-side validation

Contributors

Name	Contact	Email
Chandana GV	+91-9902477354	gchandana7799@gmail.com
Rithish Reddy	+91-9380073437	rithishreddy7007@gmail.com
Jasti SriHarsha	+91-9110342158	sri814823@gmail.com
Balaji Subhash	+91-7036739735	gbalajisubash@gmail.com
Bilakanti Varun	+91-8265739735	billannara@gmail.com
CH Mounish	+91-8374827408	k.bsairam1210@gmail.com

License

This project is created for educational purposes.

Copyright 2021 ASRS. All Rights Reserved.

Acknowledgments

- [jQuery Seat Charts](#) - Interactive seat map library
- [QRCode.js](#) - QR code generation
- [Bootstrap](#) - Frontend framework
- [jQuery](#) - JavaScript library

Support

For issues or questions, please contact the contributors listed above or raise an issue in the project repository.